

Minds and machines*

The various issues and puzzles that make up the traditional mind-body problem are wholly linguistic and logical in character: whatever few empirical 'facts' there may be in this area support one view as much as another. I do not hope to establish this contention in this paper, but I hope to do something toward rendering it more plausible. Specifically, I shall try to show that all of the issues arise in connection with any computing system capable of answering questions about its own structure, and have thus nothing to do with the unique nature (if it *is* unique) of human subjective experience.

To illustrate the sort of thing that is meant: one kind of puzzle that is sometimes discussed in connection with the 'mind-body problem' is the puzzle of *privacy*. The question 'How do I know I have a pain?' is a *deviant*† ('logically odd') question. The question 'How do I know Smith has a pain?' is not at all deviant. The difference can also be mirrored in impersonal questions: 'How does anyone ever know he himself has a pain?' is deviant; 'How does anyone ever know that someone else is in pain?' is non-deviant. I shall show that the difference in status between the last two questions is mirrored in the case of machines: if *T* is a *Turing machine* (see below), the question 'How does *T* ascertain that it is in state *A*?' is, as we shall see, 'logically odd' with a vengeance; but if *T* is capable of investigating its neighbor machine *T'* (say, *T* has electronic 'sense-organs' which 'scan' *T'*), the question 'How does *T* ascertain that *T'* is in state *A*?' is not at all odd.

Another question connected with the 'mind-body problem' is the question whether or not it is ever permissible to identify mental events and physical events. Of course, I do not claim that this question arises for Turing machines, but I do claim that it is possible to construct a logical analogue for this question that does arise, and that all of the question of 'mind-body identity' can be mirrored in terms of the analogue.

* First published in Sidney Hook (ed.) *Dimensions of Mind* (New York, 1960). Reprinted by permission of New York University Press.

† By a 'deviant' utterance is here meant one that deviates from a semantical regularity (in the appropriate natural language). The term is taken from Ziff, 1960.

To obtain such an analogue, let us identify a scientific theory with a 'partially-interpreted calculus' in the sense of Carnap†. Then we can perfectly well imagine a Turing machine which generates theories, tests them (assuming that it is possible to 'mechanize' inductive logic to some degree), and 'accepts' theories which satisfy certain criteria (e.g. predictive success). In particular, if the machine has electronic 'sense organs' which enable it to 'scan' itself while it is in operation, it may formulate theories concerning its own structure and subject them to test. Suppose the machine is in a given state (say, 'state A') when, and only when, flip-flop 36 is on. Then this statement: 'I am in state A when, and only when, flip-flop 36 is on', may be one of the theoretical principles concerning its own structure accepted by the machine. Here 'I am in state A' is, of course, 'observation language' for the machine, while 'flip-flop 36 is on' is a 'theoretical expression' which is partially interpreted in terms of 'observables' (if the machine's 'sense organs' report by printing symbols on the machine's input tape, the 'observables' in terms of which the machine would give a partial operational definition of 'flip-flop 36 being on' would be of the form 'symbol # so-and-so appearing on the input tape'). Now all of the usual considerations for and against mind-body identification can be paralleled by considerations for and against saying that state A is in fact *identical* with flip-flop 36 being on.

Corresponding to Occamist arguments for 'identify' in the one case are Occamist arguments for identity in the other. And the usual argument for dualism in the mind-body case can be paralleled in the other as follows: for the machine, 'state A' is directly observable; on the other hand, 'flip-flops' are something it knows about only via highly-sophisticated inferences – How *could* two things so different *possibly* be the same? This last argument can be put into a form which makes it appear somewhat stronger. The proposition:

(1) I am in state A if, and only if, flip-flop 36 is on,

is clearly a 'synthetic' proposition for the machine. For instance, the machine might be in state A and its sense organs might report that flip-flop 36 was *not* on. In such a case the machine would have to make a methodological 'choice' – namely, to give up (1) or to conclude that it had made an 'observational error' (just as a human scientist would be confronted with similar methodological choices in studying his own

†Cf. Carnap 1953 and 1956. This model of a scientific theory is too oversimplified to be of much general utility, in my opinion: however, the oversimplifications do not affect the present argument.

psychophysical correlations). And just as philosophers have argued from the synthetic nature of the proposition:

(2) I am in pain if, and only if, my C-fibers are stimulated,

to the conclusion that the *properties* (or 'states' or 'events') being in pain, and having C-fibers stimulated, cannot possibly be the same (otherwise (2) would be analytic, or so the argument runs); so one should be able to conclude from the fact that (1) is synthetic that the two properties (or 'states' or 'events') – being in state A and having flip-flop 36 on – cannot possibly be the same!

It is instructive to note that the traditional argument for dualism is not at all a conclusion from 'the raw data of direct experience' (as is shown by the fact that it applies just as well to non-sentient machines), but a highly complicated bit of reasoning which depends on (a) the reification of universals† (e.g. 'properties', 'states', 'events'); and on (b) a sharp analytic-synthetic distinction.

I may be accused of advocating a 'mechanistic' world-view in pressing the present analogy. If this means that I am supposed to hold that machines think,‡ on the one hand, or that human beings are machines, on the other, the charge is false. If there is some version of mechanism sophisticated enough to avoid these errors, very likely the considerations in this paper support it.§

1. Turing Machines

The present paper will require the notion of a *Turing machine*|| which will now be explained.

Briefly, a Turing machine is a device with a finite number of internal configurations, each of which involves the machine's being in one of a finite number of *states*,¶ and the machine's scanning a tape on which certain symbols appear.

† This point was made by Quine in Quine, 1957.

‡ Cf. Ziff's paper (1959) and the reply (1959) by Smart. Ziff has informed me that by a 'robot' he did not have in mind a 'learning machine' of the kind envisaged by Smart, and he would agree that the considerations brought forward in his paper would not necessarily apply to such a machine (if it can properly be classed as a 'machine' at all). On the question of whether 'this machine thinks (feels, etc.)' is *deviant* or not, it is necessary to keep in mind both the point raised by Ziff (that the important question is not whether or not the utterance is deviant, but whether or not it is deviant for non-trivial reasons), and also the 'diachronic-synchronic' distinction discussed in section 5 of the present paper.

§ In particular, I am sympathetic with the general standpoint taken by Smart in (1959b) and (1959c). However, see the linguistic considerations in section 5.

|| For further details, cf. Davis, 1958 and Kleene, 1952.

¶ This terminology is taken from Kleene, 1952, and differs from that of Davis and Turing.

will be degenerate answers – e.g. ‘By being in state *A*’ and ‘By having the pain.’

At this point it is, I believe, very clear that the difficulty has in both cases the same cause. Namely, the difficulty is occasioned by the fact that the ‘verbal report’ (‘I am in state *A*’, or ‘I am in pain’) issues directly from the state it ‘reports’: no ‘computation’ or additional ‘evidence’ is needed to arrive at the ‘answer’. And the philosophic disagreements over ‘how to talk’ are at bottom concerned with finding a terminology for describing cognitive processes in general that is not misleading in this particular case. (Note that the traditional epistemological answer to (b₁) – namely, ‘by introspection’ – is false to the facts of this case, since it clearly implies the occurrence of a mental event (the ‘act’ of introspection) distinct from the feeling of pain.)

Finally, let us suppose that the machine is equipped to ‘scan’ its neighbor machine *T*₁. Then we can see that the question ‘How does *T* ascertain that *T*₁ is in state *A*?’ may be a perfectly sensible question, as much so as ‘How does *T* ascertain that the 3000th digit of π is so-and-so?’ In both cases the answer will involve describing a whole ‘program’ (plus explaining the *rationale* of the program, if necessary). Moreover, it will be necessary to say something about the physical context linking *T* and *T*₁ (arrangement of sense organs, etc.), and not just to describe the internal states of *T*: this is so because *T* is now answering an *empirical* and not a mathematical question. In the same way ‘How did Sherlock Holmes know that Jones was in pain?’ may be a perfectly sensible question, and may have quite a complicated answer.

3. ‘Mental’ states and ‘logical’ states

Consider the two questions:

- (1) How does Jones know he has a pain?
- (2) How does Jones know he has a fever?

The first question is, as we saw in the preceding section, a somewhat peculiar one. The second question may be quite sensible. In fact, if Jones says ‘I have a pain’ no one will retort ‘You are mistaken’. (One *might* retort ‘You have made a slip of the tongue’ or ‘You are lying’, but not ‘You are *mistaken*’.) On the other hand, if Jones says ‘I have a fever’, the doctor who has just taken Jones’ temperature may quite conceivably retort ‘You are mistaken’. And the doctor need not mean that Jones made a linguistic error, or was lying, or confused.

It might be thought that, whereas the difference between statements about one’s own state and statements about the states of others has an

analogue in the case of machines, the difference, just touched upon, between statements about one's 'mental' state and statements about one's 'physical' state, in traditional parlance, does not have any analogue. But this is not so. Just what the analogue is will now be developed.

First of all, we have to go back to the notion of a Turing machine. When a Turing machine is described by means of a 'machine table', it is described as something having a tape, a printing device, a 'scanning' device (this may be no more than a point of the machine which at any given time is aligned with just one square of the tape), and a finite set (A, B, C , etc.) of 'states'. (In what follows, these will be referred to at times as *logical states* to distinguish them from certain other states to be introduced shortly.) Beyond this it is described only by giving the deterministic rules which determine the order in which the states succeed each other and what is printed when.

In particular, the 'logical description' of a Turing machine does not include any specification of the *physical nature* of these 'states' – or indeed, of the physical nature of the whole machine. (Shall it consist of electronic relays, of cardboard, of human clerks sitting at desks, or what?) In other words, a given 'Turing machine' is an *abstract* machine which may be physically realized in an almost infinite number of different ways.

As soon as a Turing machine is physically realized, however, something interesting happens. Although the machine has from the logician's point of view only the states A, B, C , etc., it has from the engineer's point of view an almost infinite number of additional 'states' (though not in the same sense of 'state' – we shall call these *structural states*). For instance, if the machine consists of vacuum tubes, one of the things that may happen is that one of its vacuum tubes may fail – this puts the machine in what is from the physicist's if not the logician's point of view a different 'state'. Again, if the machine is a manually operated one built of cardboard, one of its possible 'non-logical' or 'structural' states is obviously that its cardboard may buckle. And so on.

A physically realized Turing machine may have no way of ascertaining its own structural state, just as a human being may have no way of ascertaining the condition of his appendix at a given time. However, it is extremely convenient to give a machine electronic 'sense organs' which enable it to scan itself and to detect minor malfunctions. These 'sense organs' may be visualized as causing certain symbols to be printed on an 'input tape' which the machine 'examines' from time to time. (One minor difficulty is that the 'report' of a sense organ might occupy a number of squares of tape, whereas the machine only 'scans'

one square at a time – however, this is unimportant, since it is well known that the effect of ‘reading’ any finite number of squares can be obtained using a program which only requires one square to be scanned at a time.)

(By way of a digression, let me remark that the first actually constructed digital computers did not have any devices of the kind just envisaged. On the other hand, they *did* have over 3000 vacuum tubes, some of which were failing at any given time! The need for ‘routines’ for self-checking therefore quickly became evident.)†

A machine which is able to detect at least some of its own structural states is in a position very analogous to that of a human being, who can detect some but not all of the malfunctions of his own body, and with varying degrees of reliability. Thus, suppose the machine ‘prints out’: ‘Vacuum tube 312 has failed’. The question ‘How did the machine ascertain that vacuum tube 312 failed?’ is a perfectly sensible question. And the answer may involve a reference to both the physical structure of the machine (‘sense organs’, etc.) and the ‘logical structure’ (program for ‘reading’ and ‘interpreting’ the input tape).

If the machine prints: ‘Vacuum tube 312 has failed’ when vacuum tube 312 is in fact functioning, the mistake may be due to a miscomputation (in the course of ‘reading’ and ‘interpreting’ the input tape) or to an incorrect signal from a sense organ. On the other hand, if the machine prints: ‘I am in state A’, and it does this simply because its machine table contains the instruction: *Print: ‘I am in state A when in state A’*, then the question of a miscomputation cannot arise. Even if some accident causes the printing mechanism to print: ‘I am in state A’ when the machine is *not* in state A, there was not a ‘miscomputation’ (only, so to speak, a ‘verbal slip’).

It is interesting to note that just as there are two possible descriptions of the behavior of a Turing machine – the engineer’s structural blueprint and the logician’s ‘machine table’ – so there are two possible descriptions of human psychology. The ‘behavioristic’ approach (including in this category theories which employ ‘hypothetical constructs’, including ‘constructs’ taken from physiology) aims at eventually providing a complete physicalistic‡ description of human behavior, in terms which link up with chemistry and physics. This corresponds to the engineer’s or physicist’s description of a physically realized Turing

† Actually, it was not necessary to add any ‘sense organs’; existing computers check themselves by ‘performing crucial experiments with themselves’ (i.e. carrying out certain test computations and comparing the results with the correct results which have been given).

‡ In the sense of Oppenheim, 1958; not in the ‘epistemological’ sense associated with Carnap’s writing on ‘physicalism’.

machine. But it would also be possible to seek a more abstract description of human mental processes, in terms of 'mental states' (physical realization, if any, unspecified) and 'impressions' (these play the role of symbols on the machine's tapes) – a description which would specify the laws controlling the order in which the states succeeded one another, and the relation to verbalization (or, at any rate, verbalized thought). This description, which would be the analogue of a 'machine table', it was in fact the program of classical psychology to provide! Classical psychology is often thought to have failed for *methodological* reasons; I would suggest, in the light of this analogy, that it failed rather for empirical reasons – the mental states and 'impressions' of human beings do not form a causally closed system to the extent to which the 'configurations' of a Turing machine do.

The analogy which has been presented between logical states of a Turing machine and mental states of a human being, on the one hand, and structural states of a Turing machine and physical states of a human being, on the other, is one that I find very suggestive. In particular, further exploration of this analogy may make it possible to further clarify the notion of a 'mental state' that we have been discussing. This 'further exploration' has not yet been undertaken, at any rate by me, but I should like to put down, for those who may be interested, a few of the features that seem to distinguish logical and mental states respectively from structural and physical ones:

(1) The functional organization (problem solving, thinking) of the human being or machine can be described in terms of the sequences of mental or logical states respectively (and the accompanying verbalizations), without reference to the nature of the 'physical realization' of these states.

(2) The states seem intimately connected with *verbalization*.

(3) In the case of rational thought (or computing), the 'program' which determines which states follow which, etc., is open to rational criticism.

4. Mind-body 'identity'

The last area in which we have to compare human beings and machines involves the question of *identifying* mental states with the corresponding physical states (or logical states with the corresponding structural states). As indicated at the beginning of this paper, all of the arguments for and against such identification can perfectly well be discussed in terms of Turing machines.